

An aerial photograph of Madrid, Spain, taken at sunset. The sky is a mix of orange, pink, and blue. The city below is densely packed with buildings, many of which are illuminated with warm lights. The Goya statue is visible in the center of the image, atop a building. The overall scene is a vibrant and detailed representation of the city's architecture and atmosphere.

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Programación Python

**Estructuras, Elementos, Funciones
y Tipos de datos básicos y Python
Debugging**

Python function challenge.

Find the pattern, then write a function: `soco_bate_vira(n)`

That receives:

```
> soco_bate_vira(2)
```

```
[1] Soco Soco Bate Bate Soco Soco Vira Vira
```

```
> soco_bate_vira(1)
```

```
[1] Soco Bate Soco Vira
```

```
> soco_bate_vira(0)
```

```
[1] Soco Bate Vira
```

What is the result of `soco_bate_vira(3)`?

```
soco_bate_vira(4)
```

```
soco_bate_vira(5)
```

```
soco_bate_vira(N)
```

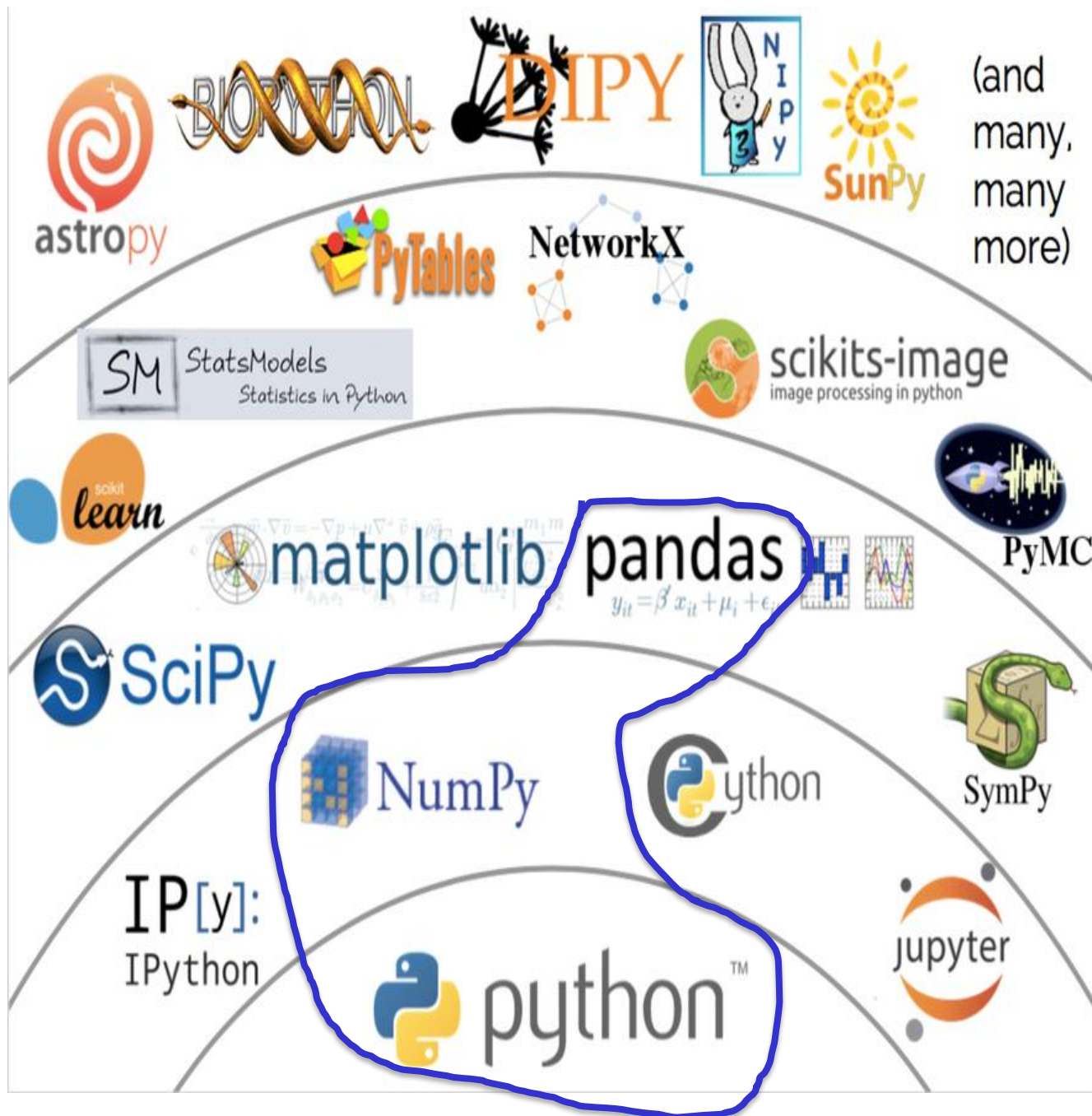
```
soco_bate_vira(100)
```

https://www.youtube.com/watch?v=VZGcm_JYOKM

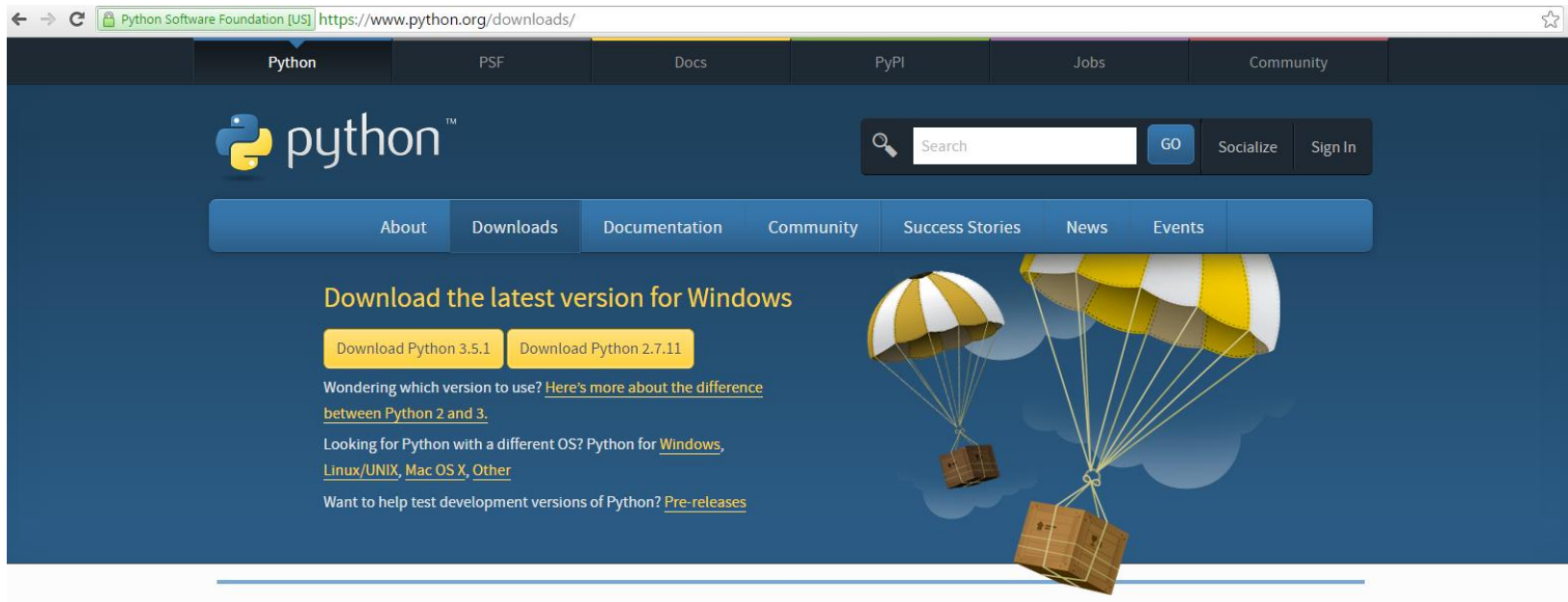


Chocolate prize next F2F class for anyone that can do this version in class:

https://www.youtube.com/watch?v=zK_3onMpwwU



Hello World!



```
print("hello World")
```

It is Kahoot time



Kahoot!

[Python Basics](#)


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CALM
IT'S
KAHOOT
TIME**

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Functions

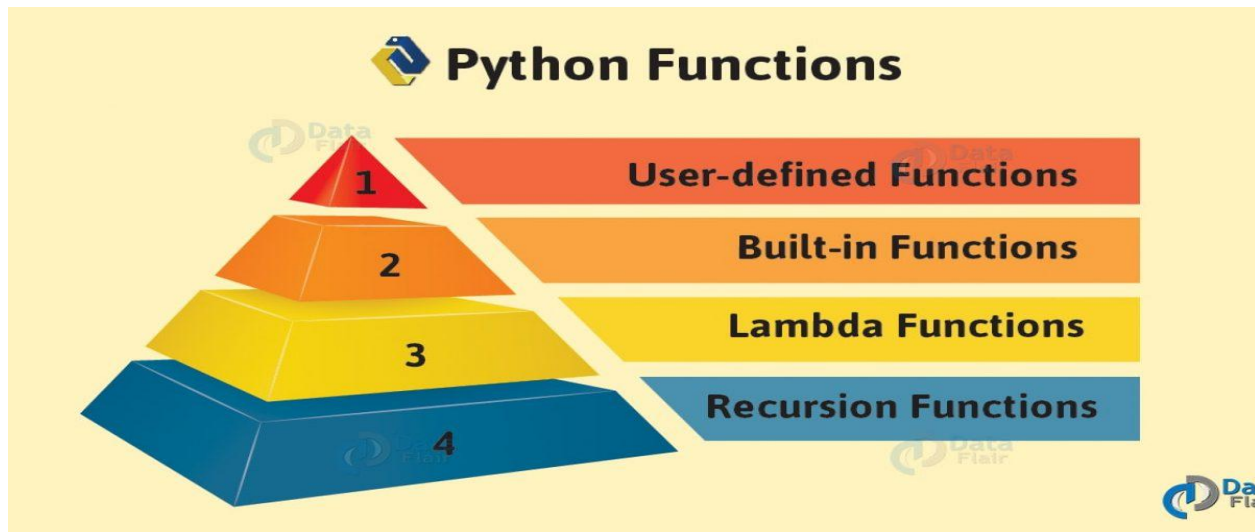
a named section of a program
that performs a specific task

Python Functions



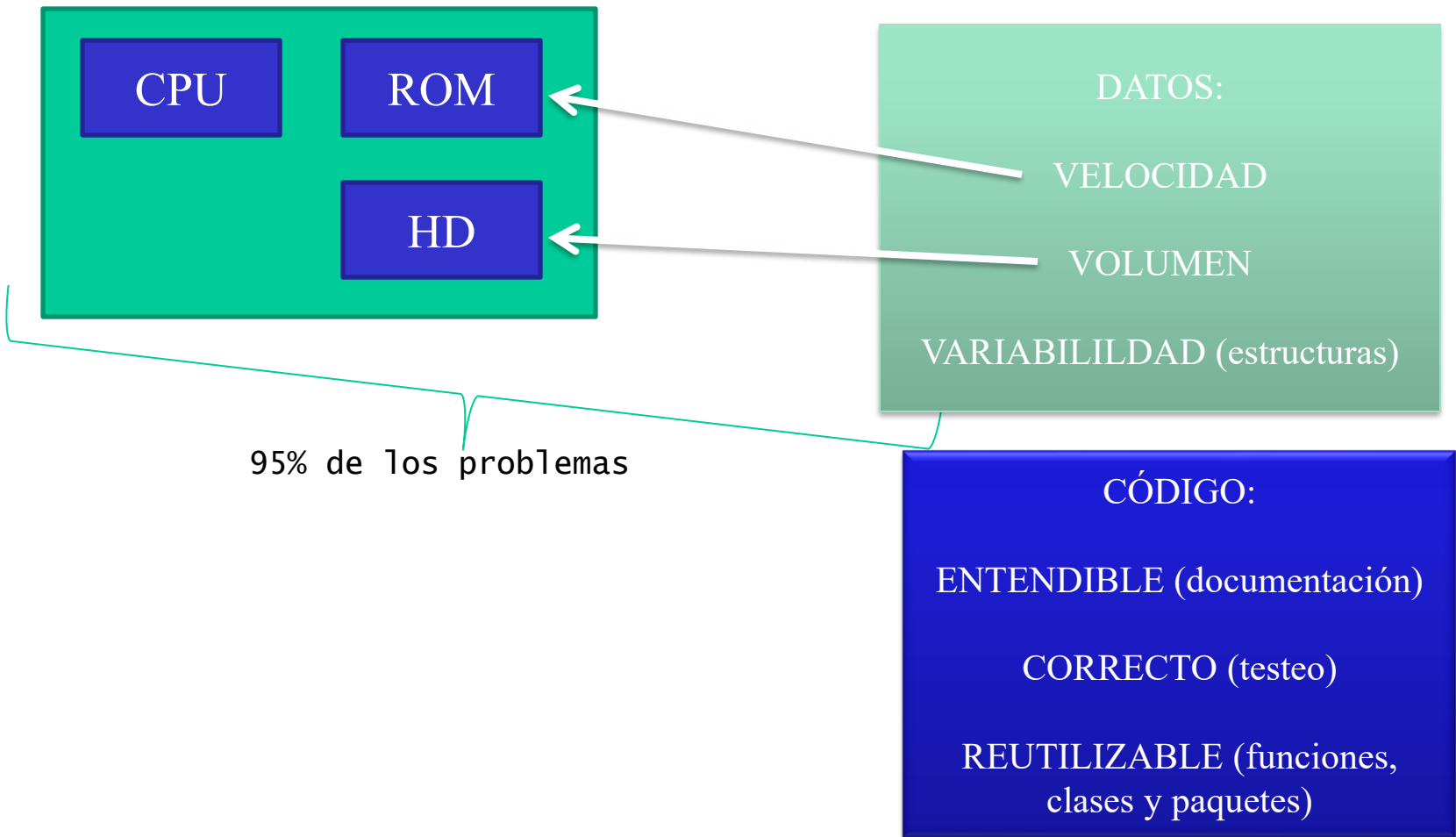
Functions

a named section of a program that performs a specific task



Functions:

A way of organizing code to make it more understandable, easy to use, reproducible and easier to share.





SCHOOL OF
HUMAN SCIENCES
& TECHNOLOGY

```
#### FUNCTIONS AND NOTHING - None is a type in python
#### FUNCIONES Y LA NADA - None es un tipo en python
def func1():
    """ this is my own doc string - it becomes available to anyone who calls help(func1). """
    pass

if func1()==None:
    print("It returned the 'nothing'! / Me ha devuelto la 'nada'!")
else:
    print("It returned something!")

help(func1)
```

```
#### FUNCTIONS AND NOTHING - None is a type in python
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def func1():
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help(func1)
```

Numeric
Strings
Constructors
Iterables (Lists, Dicts,...)

Built-in Functions

Python Functioning
Input / Output

abs() – Numeric

delattr()

hash()

memoryview()

set() - Constructors

all()

dict() - Constructors

help()

min() - Numeric

setattr()

any()

dir()

hex() - Constructors

next()

slice()

ascii()

divmod()

id()

object()

sorted()

bin()

enumerate()

input() - Input / Output

oct()

staticmethod()

bool() - Constructors

eval() - Python Functioning

int()

open() - Input / Output

str()

breakpoint()

exec()

isinstance()

ord()

sum()

bytearray()

filter()

issubclass()

pow()

super()

bytes() - Constructors

float()

iter()

print() - Input / Output

tuple()

callable()

format() - Strings

len()

property()

type()

chr() - Strings

frozenset()

list() - Constructors

range() - Iterables

vars()

classmethod()

getattr()

locals()

repr()

zip()

compile() - Python
Functioning

globals()

map()

reversed() - Strings

__import__()

complex() - Constructors

hasattr()

max()

<https://docs.python.org/3/library/functions.html>

Python Basic Exercises 3 - function

Write a function which can compute the factorial of a given numbers. Suppose the following input is supplied to the program: 8

Then, the output should be: 40320

Hint: `x=int(input("Enter the n to obtain the factorial(n):"))`

Python Basic Exercises 4 - function

- **Write a program that invokes a function that calculates and returns the value according to the given formula:**

$$Q = [(2 * C * D)/H]$$

Call your function several times, using integer values of your choice to C, D and H. The output of your function should always be integer.

Hint: If the output received is in decimal form, it should be rounded off to its nearest value (for example, if the output received is 26.1, it should be printed as 26)

It is Kahoot time

The Kahoot! logo is displayed in a large, white, sans-serif font. It is centered over a background that is split into four quadrants: red (top-left), blue (top-right), yellow (bottom-left), and green (bottom-right). Each quadrant contains a faint, stylized map of a world region.

Python Arithmetic's &
Functions

A purple rectangular poster with a white crown icon at the top. Below the crown, the text "KEEP CALM IT'S KAHOOT TIME" is written in white, bold, sans-serif capital letters. At the bottom right, the website "KeepCalmAndPosters.com" is written in a small white font.

KEEP
CALM
IT'S
KAHOOT
TIME

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Lists, tuples and Dictionaries

Standard Data Types:

Python has five standard data types:

- Numbers
- String
- List
- Tuple
- Dictionary

Python Lists



Python Basic Exercises 5 - Lists

- **Write a program which accepts a sequence of comma-separated numbers from console and generate a list and a tuple with the square of every number.**
- **Suppose the following input is supplied to the program:**
34,67,55,33,12,98
- **Then, the output should be:**
[1156, 4489, 3025, 1089, 144, 9604]
(1156, 4489, 3025, 1089, 144, 9604)

Hint: In case of input data being supplied to the question, it should be assumed to be a console input. tuple() constructor method can convert list to tuple

Python Lists:

- Lists are the most versatile of Python's compound data types. A list contains items separated by commas and enclosed within square brackets ([]).
- To some extent, lists are similar to arrays in C. One difference between them is that all the items belonging to a list can be of different data type.
- The values stored in a list can be accessed using the slice operator ([] and [:]) with indexes starting at 0 in the beginning of the list and working their way to end-1.
- The plus (+) sign is the list concatenation operator, and the asterisk (*) is the repetition operator.

```
list = [ 'abcd', 786 , 2.23, 'john', 70.2 ]  
tinylist = [123, 'john']
```

```
print list      # Prints complete list  
print list[0]   # Prints first element of the list  
print list[1:3] # Prints elements starting from 2nd till 3rd  
print list[2:]  # Prints elements starting from 3rd element  
print tinylist * 2 # Prints list two times  
print list + tinylist # Prints concatenated lists
```

Output:

```
['abcd', 786, 2.23, 'john', 70.2]  
abcd  
[786, 2.23]  
[2.23, 'john', 70.2]  
[123, 'john', 123, 'john']  
['abcd', 786, 2.23, 'john', 70.2, 123, 'john']
```

Python Tuples:

- A tuple is another sequence data type that is similar to the list. A tuple consists of a number of values separated by commas. Unlike lists, however, tuples are enclosed within parentheses.
- The main differences between lists and tuples are: Lists are enclosed in brackets ([]), and their elements and size can be changed, while tuples are enclosed in parentheses (()) and cannot be updated. Tuples can be thought of as **read-only** lists.

```
tuple = ( 'abcd', 786 , 2.23, 'john', 70.2 )
```

```
tinytuple = (123, 'john')
```

```
print tuple      # Prints complete list
```

```
print tuple[0]   # Prints first element of the list
```

```
print tuple[1:3] # Prints elements starting from 2nd till 3rd
```

```
print tuple[2:]  # Prints elements starting from 3rd element
```

```
print tinytuple * 2 # Prints list two times
```

```
print tuple + tinytuple # Prints concatenated lists
```

OUTPUT:

```
('abcd', 786, 2.23, 'john', 70.2)
```

```
abcd
```

```
(786, 2.23)
```

```
(2.23, 'john', 70.2)
```

```
(123, 'john', 123, 'john')
```

```
('abcd', 786, 2.23, 'john', 70.2, 123, 'john')
```

Python Dictionaries



Python Basic Exercises 6 - Dictionary

- **With a given integral number n , write a program to generate a dictionary that contains $(i, i*i)$ such that i is an integral number between 1 and n (both included). Finally, the program should print the dictionary.**
- **Suppose the following input is supplied to the program: 8
Then, the output should be:
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64}**

Hint: In case of input data being supplied to the question, it should be assumed to be a console input. Consider using the constructor method `dict()`

Python Dictionary:

- Python 's dictionaries are hash table type. They work like associative arrays or hashes found in Perl and consist of key-value pairs.
- Keys can be almost any Python type, but are usually numbers or strings. Values, on the other hand, can be any arbitrary Python object.
- Dictionaries are enclosed by curly braces ({ }) and values can be assigned and accessed using square braces ([]).

```
dict = {}  
dict['one'] = "This is one"  
dict[2] = "This is two"  
tinydict = {'name': 'john', 'code': 6734, 'dept': 'sales'}  
print dict['one']    # Prints value for 'one' key  
print dict[2]        # Prints value for 2 key  
print tinydict       # Prints complete dictionary  
print tinydict.keys() # Prints all the keys  
print tinydict.values() # Prints all the values
```

OUTPUT:

This is one

This is two

{'dept': 'sales', 'code': 6734, 'name': 'john'}

['dept', 'code', 'name']

['sales', 6734, 'john']

50%
Good / Appropriate
Algorithm

30%
Good / Appropriate
Data Structure

20%
Faster / +Memory
Computer

Like Constants!

list

tuple

dictionary

set

COMMON:

DIFFERENTIAL: *Mutable, Ordered [0,n-1]*

CREATE EMPTY: `list1=list(); list2=[]`

CREATE: `list1 = [3,6,7,3,7]`

INSERT: `list1.append(8)`

INSERT IN POSITION: `list1.insert(3,'value')`

REMOVE: `list1.pop(1)`
`/ list1.remove(3)`

Collection of 0, 1 or many of any combination of types of Python scalars

Immutable, Ordered [0,n-1]

`tuple1=tuple(); tuple2=()`

`tupple1 = (3,3,6,7,7)`

-

-

-

*Key is Unique and Unordered
(key immutable / Value = Mutable)*

`dict1=dict(); dict2={}`

`dict1 = {3:'str',6:False}`

`dict1['Juan'] = 'Casado'`

-

`del dict1['Juan']`

Unique, Unordered

`set1 = set()`

`set1 = {3,6,7}`

`set1.add(8)`

-

`set1.remove(8)`

Data frame (pandas)

Key-Value (pickledb)

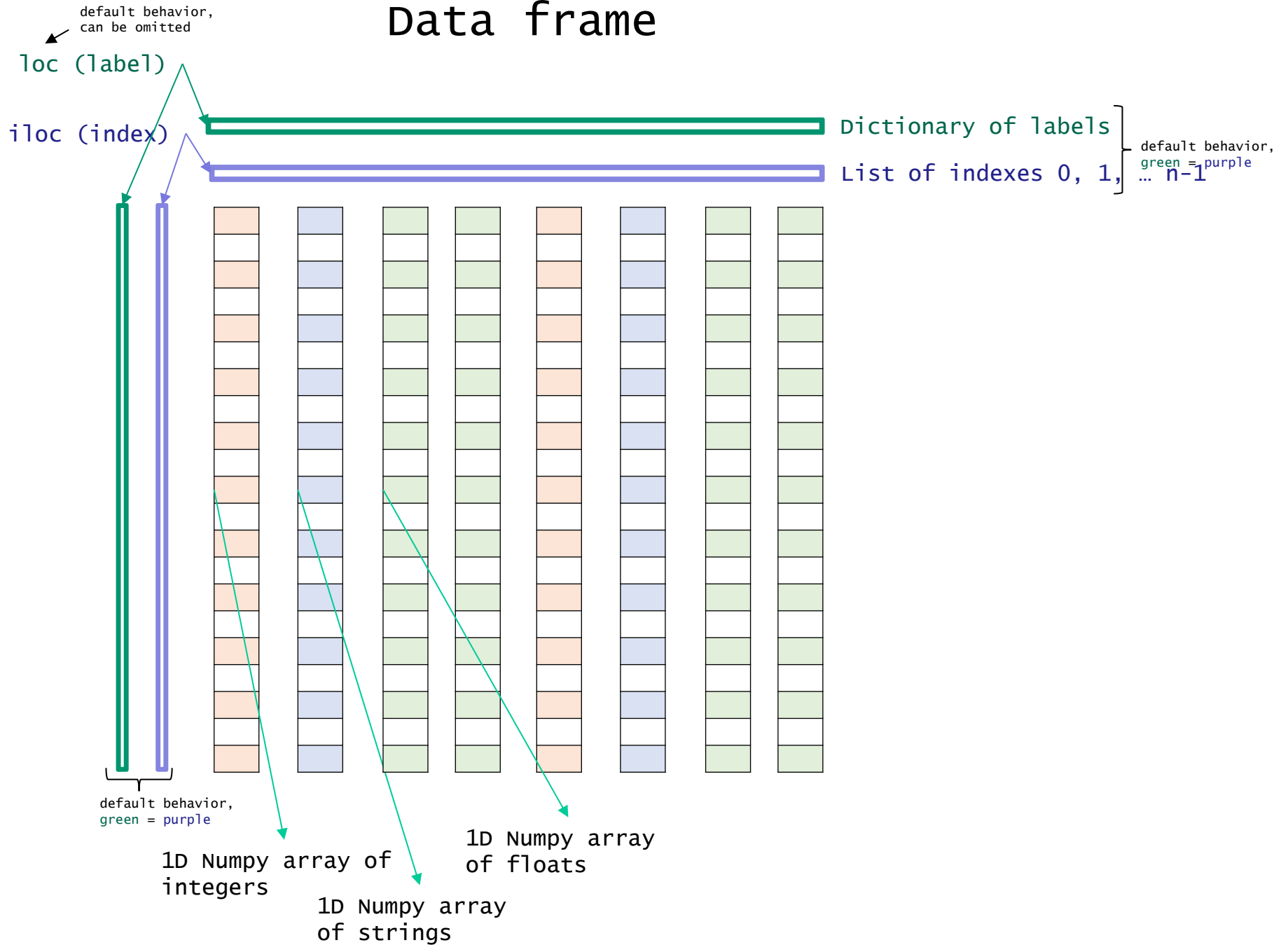
HDFS (pytables)

SQL

Pure Python
(comes within
the language)

Python
Ecosystem
(libraries)

Data frame



It is Kahoot time



Kahoot!

Python Lists & Dictionaries
- Session 5 & 6



Data frame

